PROJECT PROPOSAL

Investment Decision Data Platform

DEPI • Data Analytics / Data Engineering Project Proposal

1. Objective

To combine gold prices, exchange rates, and selected economic indicators in one organized data platform, so users can explore market trends and make more informed investment decisions.

2. Project Description

The project proposes a data workflow for collecting, cleaning, organizing, and analyzing financial and economic data related to gold. It will explore how gold prices move alongside exchange rates and indicators such as inflation, interest rates, Treasury yields, and the U.S. Dollar Index. The analysis may include price trends, returns, volatility, and correlations. Results are intended to support analysis, not to guarantee profits or provide personalized financial advice.

3. Tools & Design

Proposed workflow: Data Sources → Data Collection → Data Cleaning & Transformation → Data Storage → Financial Analysis → Dashboard / Visualizations. The design will be refined after the team reviews the available data and confirms the project requirements.

4. Technologies to Use

Python, Pandas, NumPy, SQL, CSV files and suitable public APIs/data downloads; PostgreSQL or MySQL for storage; and Power BI or Tableau for visualizations. The final tool choices will depend on data availability and project requirements.

5. Team Members

01	Lujaina Ahmed
02	Mona Islam
03	Martina Emad
04	Rawan Othman
05	Sama Mohamed
06	Nourhan Ahmed
07	Ziad Ashraf

6. Week 1 — Planning & Requirements

Target date: 16 October 2026

Week 1 focuses on planning, reviewing related work, and gathering requirements. Each team member has one clear contribution, and the findings will be combined into the shared proposal.

- **Mona Islam:** Review two or three related financial-data projects and summarize their goals, useful features, and ideas we could adapt.
- **Martina Emad:** Research possible sources for historical gold prices and record each source link, date range, and available format.
- **Rawan Othman:** Research the economic indicators relevant to gold prices—exchange rates, inflation, interest rates, Treasury yields, and the U.S. Dollar Index—and briefly explain why each may matter.
- **Sama Mohamed:** Prepare a practical data-quality checklist covering missing values, duplicates, date formats, inconsistent units, and data reliability.
- **Nourhan Ahmed:** Define the problem statement, target users, project objectives, and key questions the platform should help answer; draft the project scope and proposed workflow.
- **Ziad Ashraf:** Draft the dashboard requirements and recommend suitable charts or summary indicators for presenting trends and comparisons clearly.
- **Lujaina Ahmed:** Define the proposed financial analyses—price trends, returns, volatility, and correlations—and write one question each analysis could help answer.

Week 1 Deliverables

- A problem statement, target-user description, project objectives, and initial scope.
- A short review of similar projects and their useful features.
- An initial list of gold-price sources, economic indicators, and required data fields.
- Initial data-quality, financial-analysis, and dashboard requirements, plus a proposed workflow.

7. Week 2 — System Analysis & Design

Target date: 6 November 2026

Tasks:

• Nourhan Ahmed — Finalize the problem statement, objectives, users, and project scope based on Week 1 findings.

• Martina Emad and Rawan Othman — Confirm the selected data sources and document the required data fields and indicators.

• Sama Mohamed — Define the data-cleaning and validation rules that will be applied before analysis.

• Lujaina Ahmed — Specify the financial calculations and analysis outputs, including trends, returns, volatility, and correlations.

• Ziad Ashraf and Mona Islam — Prepare the system workflow and a simple dashboard layout; the team reviews the design together.

Deliverables

• Requirements document (functional and non-functional requirements).

• System workflow / data pipeline diagram and preliminary database design.

• Initial dashboard wireframe and agreed technology choices.

8. Week 3 — Implementation

Target date: 30 November 2026

Tasks:

• Martina Emad — Collect a small, suitable sample of historical gold-price data and document its source.

• Rawan Othman — Collect available economic-indicator data and align the relevant dates and units.

• Sama Mohamed — Clean the collected data and check missing values, duplicates, formats, and consistency.

• Lujaina Ahmed — Implement and test the planned financial analyses using the prepared data.

• Ziad Ashraf — Build the first dashboard visuals using the agreed charts and indicators.

• **Mona Islam and Nourhan Ahmed —** Check that the implementation matches the project requirements and keep the project documentation updated.

Deliverables

• Collected and cleaned dataset(s) with source notes.

• Working initial analysis and dashboard prototype.

• Updated project documentation and implementation notes.

9. Week 4 — Testing, Documentation & Presentation Preparation

Target date: 4 December 2026

Tasks:

• Sama Mohamed — Test data quality and record any remaining data issues.

• Lujaina Ahmed and Rawan Othman — Verify the financial calculations and interpret the main results carefully.

• Ziad Ashraf — Refine dashboard labels, chart readability, and presentation layout.

• Martina Emad — Verify source links, data coverage, and dataset descriptions.

• Mona Islam — Prepare the literature-review summary and help organize the final report.

• **Nourhan Ahmed —** Review the problem, objectives, and requirements against the final result; coordinate the presentation outline with the team.

Deliverables

• Tested project prototype and a short list of known limitations.

• Final report with sources, methodology, analysis, and results.

• Presentation slides and a clear demonstration plan.